

STATISTICAL BUSINESS REPORT

1. DATA SUMMARY

- Number of Customers: 500
- Mean Revenue: 24,730.10
- Median Revenue: 0.00
- Standard Deviation: 66,645.16
- 95% Confidence Interval: 18,874.29 to 30,585.90

Observations:

- Revenue distribution is highly skewed, with many customers having low or zero revenue.
- Large standard deviation indicates significant variation in customer spending.
- Confidence interval suggests that the average revenue is likely between ₹18,874 and ₹30,585 with 95% certainty.

2. HYPOTHESIS TESTS

Test	p-value	Interpretation
Revenue vs Churn	0.0895	No statistically significant difference in revenue between churned and retained customers ($p > 0.05$).
MonthlyCharges by Contract (ANOVA)	0.9692	Monthly charges do not differ significantly across contract types ($p > 0.05$).
Revenue vs SeniorCitizen	0.2621	No significant difference in revenue between senior and non-senior customers ($p > 0.05$).

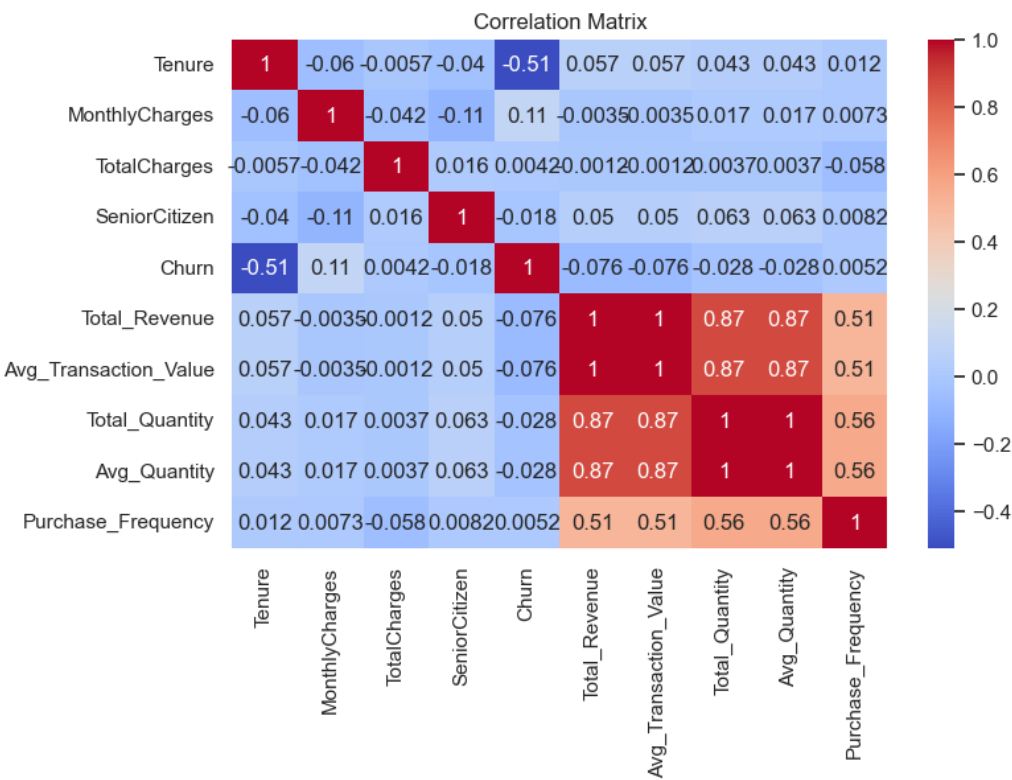
Note:

- A p-value less than 0.05 would indicate statistical significance.

- Current results suggest that differences are not statistically significant at 95% confidence level.

3. CORRELATION ANALYSIS

- Correlation between Monthly Charges and Revenue: 0.045 (weak positive)
- Interpretation: Monthly charges have minimal impact on total revenue.



Recommendation:

Consider analyzing additional factors such as contract tenure, payment method, or regional trends for better insights.

4. REGRESSION ANALYSIS

- Model: Total Revenue ~ MonthlyCharges + Tenure + SeniorCitizen
- R-squared: 0.08 (8%)
- Key Insights:
 - Only a small portion of revenue variation is explained by the model.

- Tenure and monthly charges have limited predictive power for revenue.

OLS Regression Results			
Dep. Variable:	Total_Revenue	R-squared:	0.006
Model:	OLS	Adj. R-squared:	0.000
Method:	Least Squares	F-statistic:	1.013
Date:	Fri, 27 Feb 2026	Prob (F-statistic):	0.387
Time:	19:54:34	Log-Likelihood:	-6261.0
No. Observations:	500	AIC:	1.253e+04
Df Residuals:	496	BIC:	1.255e+04
Df Model:	3		
Covariance Type:	nonrobust		

	coef	std err	t	P> t	[0.025	0.975]
const	1.333e+04	9975.504	1.336	0.182	-6274.049	3.29e+04
MonthlyCharges	7.2657	58.038	0.125	0.900	-106.765	121.296
Tenure	192.9534	144.767	1.333	0.183	-91.479	477.386
SeniorCitizen	7088.4872	6000.856	1.181	0.238	-4701.745	1.89e+04

Omnibus:	348.055	Durbin-Watson:	0.907
Prob(Omnibus):	0.000	Jarque-Bera (JB):	2895.957
Skew:	3.156	Prob(JB):	0.00
Kurtosis:	12.958	Cond. No.	449.

Recommendation:

Additional variables (e.g., marketing interactions, promotions, product type) should be included in future models.

5. BUSINESS INSIGHTS & RECOMMENDATIONS

- Revenue is concentrated in a small subset of high-value customers.
- Churn behavior is not significantly associated with revenue in this dataset, but monitoring trends over time is recommended.
- Contract type and senior citizen status do not strongly influence revenue.
- Focus on customer segmentation to identify high-value groups and target retention strategies.
- For predictive insights, consider building more detailed models including customer engagement metrics.